

Test of Mathematics for University Admission
Mock Set F Paper 2
 Mathematical Reasoning

Name: _____

Date sat: _____

Score: _____ / 20

Answer all 20 questions. Each question has exactly one correct option and is worth one mark. No marks are deducted for wrong answers. Throughout, “must be true” means true in every case allowed by the information given, and a counterexample must satisfy the hypothesis while failing the conclusion.

[illegible]

Question 1

A , B and C are statements. You are told:

“ A only if B ”, “ C if B ”, “not C unless A ”.

Which of the following follows?

- A** A and B are equivalent, but C need not be equivalent to either.
- B** A implies C , but not conversely.
- C** A , B and C are all equivalent.
- D** C implies A , and nothing else follows.
- E** No two of A , B , C need be equivalent.
- F** B and C are equivalent, but A need not be equivalent to either.

Question 2

Which of the following statements are true?

- I** 2^{2024} ends in the digit 6.
- II** 3^{2026} ends in the digit 9.
- III** 7^{2025} ends in the digit 7.
- IV** 4^{2025} ends in the digit 6.
- A** I and II only
- B** I, II, III and IV
- C** I, II and III only
- D** II and III only
- E** I and IV only
- F** III only

Question 3

A student proves that $\left|x + \frac{1}{x}\right| \geq 2$ for every non-zero real x .

- (I) If $x > 0$: $(x - 1)^2 \geq 0$, so $x^2 + 1 \geq 2x$; dividing by $x > 0$ gives $x + \frac{1}{x} \geq 2$.
- (II) If $x < 0$: $(x + 1)^2 \geq 0$, so $x^2 + 1 \geq -2x$; dividing by $x < 0$ reverses the inequality, giving $x + \frac{1}{x} \leq -2$.
- (III) In either case, $\left|x + \frac{1}{x}\right| \geq 2$.

Which of the following is correct?

- A** The proof is valid.
- B** Line II is wrong: the inequality should not be reversed.
- C** Line III does not follow from lines I and II.
- D** Line I is wrong: it needs $x \geq 1$.
- E** The claim is false for $-1 < x < 0$.
- F** Line II uses the wrong square.

Question 4

Let S be the statement “if n is divisible by 4, then n^2 is divisible by 8”, where n is a positive integer. How many of the following are true: S , its converse, its inverse, and its contrapositive?

- A 0
- B 1
- C 2
- D 3
- E 4

Question 5

Which of the following conditions on the real number x are sufficient for $x^3 < x$?

- I $0 < x < 1$
 - II $x < 1$
 - III $x^2 < 1$
 - IV $x < -1$
- A I only
 - B I and III only
 - C IV only
 - D I and IV only
 - E I, III and IV only
 - F II only

Question 6

Call a pair (a, b) of positive integers *friendly* if $a + b$ divides ab . Which of the following statements are true?

- I If (a, b) is friendly, then so is (ka, kb) for every positive integer k .
 - II (a, a) is friendly if and only if a is even.
 - III $(1, b)$ is never friendly.
 - IV If (a, b) is friendly, then one of a and b divides the other.
- A I only
 - B I and II only
 - C I, II, III and IV
 - D I, II and III only
 - E II and III only
 - F I and IV only

Question 7

A student proves that there are infinitely many primes: “Suppose there are finitely many, $p_1, p_2, \dots, p_k$. Let $N = p_1 p_2 \cdots p_k + 1$. None of the p_i divides N . Hence N is prime; but N is larger than every p_i , a contradiction.” Which of the following is correct?

- A** The proof is correct as written.
- B** N need not be prime, but it has a prime factor not in the list, so the proof is repaired by changing that step.
- C** N need not be prime, and the argument cannot be repaired.
- D** The error is that some p_i may divide N .
- E** The argument shows only that there are more than k primes for one particular k .
- F** The assumption that the primes can be listed is itself invalid.

Question 8

Positive integers a and b have highest common factor 1. Which of the following must be true?

- I** $\gcd(a + b, ab) = 1$
- II** $\gcd(a + b, a - b)$ is 1 or 2 (when $a \neq b$)
- III** $\gcd(a^2, b^2) = 1$
- IV** $\gcd(a + b, a^2 + b^2) = 1$
- A** I, II and III only
- B** I and III only
- C** I, II, III and IV
- D** II and III only
- E** I only
- F** III only

Question 9

Which of the following statements are true?

- I** If a is rational and b is irrational, then $a + b$ is irrational.
- II** If a is rational and b is irrational, then ab is irrational.
- III** If x and y are irrational, then $x + y$ is irrational.
- IV** If x^2 is irrational, then x is irrational.
- A** I only
- B** I, II and IV only
- C** IV only
- D** I, II, III and IV
- E** I and II only
- F** I and IV only

Question 10

On an island, knights always tell the truth and knaves always lie. Three islanders make these statements:

A: "B is a knave."

B: "Exactly one of us three is a knight."

C: "B and I are different types."

What are A, B and C?

- A All three are knights.
- B A and C are knights; B is a knave.
- C Only A is a knight.
- D Only C is a knight.
- E A and B are knights; C is a knave.
- F No assignment is consistent.

Question 11

Call a function f , defined for $x \geq 0$, *subadditive* if $f(x + y) \leq f(x) + f(y)$ for all $x, y \geq 0$. Which of the following functions are subadditive?

- I $f(x) = \sqrt{x}$
- II $f(x) = x^2$
- III $f(x) = x + 1$
- IV $f(x) = 2^x$
- V $f(x) = 3x$
- A I and III only
- B I, III and V only
- C I only
- D III and V only
- E I, II and V only
- F I, II, III, IV and V

Question 12

Which of the following conditions guarantee that a triangle is equilateral?

- I Two of its angles are 60° .
- II Its angles form an arithmetic progression.
- III Its sides form an arithmetic progression and one of its angles is 60° .
- IV Its angles form an arithmetic progression and so do its sides.
- A I only
- B I and III only
- C I, III and IV only
- D I and IV only
- E I, II, III and IV
- F III and IV only

Question 13

f is a polynomial with $f(x) \geq 0$ for $0 \leq x \leq 2$ and $\int_0^2 f(x) dx = 0$. Which of the following must be true?

- I $f(1) = 0$
 - II $f(x) = 0$ for every real x .
 - III $\int_0^3 f(x) dx = 0$
 - IV The statements above would still all be true if f were only known to be continuous.
- A I only
 - B I and III only
 - C I, II, III and IV
 - D I and II only
 - E II and III only
 - F I, II and III only

Question 14

A positive integer $n \leq 100$ is the subject of five statements:

- (1) n is a perfect square. (2) n is a perfect cube. (3) n is even. (4) $n < 50$. (5) n is divisible by 9.

Exactly one of the statements is false. Find n .

- A 16
- B 18
- C 27
- D 36
- E 64
- F 81

Question 15

The cubic function f satisfies $f(0) = 0$ and $f'(0) > 0$, and it has a local maximum at $x = 2$. Which of the following must be true?

- I f has a local minimum at some $x > 2$.
 - II $f(2) > 0$
 - III $f'(1) > 0$
 - IV $f(x) = 0$ has three distinct real roots.
- A II only
 - B II and III only
 - C I, II and III only
 - D I, II, III and IV
 - E I and II only
 - F III and IV only

Question 16

Consider the equation $\tan x = x$, with x in radians. Which of the following are true?

- I** It has exactly one solution in the interval $\frac{\pi}{2} < x < \frac{3\pi}{2}$.
- II** It has no solution in the interval $0 < x < \frac{\pi}{2}$.
- III** Its solution in $\frac{\pi}{2} < x < \frac{3\pi}{2}$ is greater than π .
- IV** Its solution in $\frac{\pi}{2} < x < \frac{3\pi}{2}$ is less than $\frac{5\pi}{4}$.
- A** I, II, III and IV
- B** I only
- C** I and III only
- D** II and IV only
- E** I, II and III only
- F** I and II only

Question 17

Let $p = \log_2 3$, $q = \log_3 5$ and $r = \log_5 8$. Which of the following is correct?

- A** $r < q < p$
- B** $p < q < r$
- C** $q < r < p$
- D** $r < p < q$
- E** $q < p < r$
- F** $p < r < q$

Question 18

A function f satisfies $f(4 - x) = f(x)$ and $f(-x) = -f(x)$ for every real x . Which of the following must be true?

- I** $f(4) = 0$
- II** $f(x + 8) = f(x)$ for every x
- III** $f(2) = 0$
- IV** $f(6) = -f(2)$
- A** I and II only
- B** II only
- C** I, II, III and IV
- D** I, II and IV only
- E** II and IV only
- F** I and IV only

Question 19

For an integer k with $-10 \leq k \leq 10$, let P be the statement “ $x^2 - kx + 1 = 0$ has real roots” and Q the statement “ $x^2 - x + k = 0$ has real roots”. For how many such k is exactly one of P and Q true?

- A** 5
- B** 7
- C** 9
- D** 10
- E** 11
- F** 13

Question 20

Real numbers a , b and c satisfy $a + b + c = 0$ and $a^2 + b^2 + c^2 = 6$. Which of the following must be true?

- I** $ab + bc + ca = -3$
 - II** Each of a , b and c lies in the interval $[-2, 2]$.
 - III** $abc \leq 1$
 - IV** $a^4 + b^4 + c^4 = 18$
- A** I and II only
 - B** I, II, III and IV
 - C** I and IV only
 - D** II and IV only
 - E** I, II and IV only
 - F** I, II and III only

END OF PAPER